

Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT): A Fruit-Fly Connectome Toy Model

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Abstract

We model biological transitions as multi-agent realisations of the composite operator pipeline $G = U \circ F \circ K \circ C$ from Topographical Orthogonal Generative Theory (TO/TOGT). Using the fixed-point, contraction-mapping, and saturated-pitchfork theorems proved in the companion paper (Zenodo 10.5281/zenodo.19210137), we construct a minimal toy model of the *Drosophila* connectome. Neural clusters are treated as coupled agents whose trajectories stabilise under six applications of G . The model reproduces qualitative features of neural coherence, oscillatory entrainment, and stress-response switching (HPA-axis analogue) while remaining strictly within what is rigorously established. The dm^3 canonical invariants $(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$ are used solely as convenient normalisation choices.

MSC codes: 37C25, 37D10, 37G10, 47H10, 92B20

1 Relation to Prior Work

This paper supplements the author’s earlier preprint *Principia Orthogona, Volume One: The Mathematics of Generative Transitions* (HAL hal-05555216v1) and its companion *Principia Orthogona, Volume Two: Contact Realization of Generative Transitions* (HAL hal-05559997v1). The rigorous fixed-point theory, contraction mapping, saturated pitchfork bifurcation, and algebraic separation theorem are taken from the newly deposited standalone paper *Fixed-Point Theory and Operator Algebra for Generative Transitions* (Zenodo 10.5281/zenodo.19210137, HAL hal-05565024v1).

The complete AXLE formal verification repository (Lean 4 / Mathlib implementation of the operator pipeline, regeneration hierarchy, and bio-swarm theorems) is publicly available at <https://github.com/TOTOGT/AXLE>. The same operator pipeline and theorems underpin the architectural application *The G6 Crystal: A dm^3 -Derived Architectural Form for Resonance-Stable Tall Structures* (Zenodo 19162013) and the explicit toy-model analysis *The dm^3 Operator: Explicit Toy Model and Global Dynamical Analysis* (Zenodo 19117400).

2 The Generative Operator Pipeline

Definition 2.1. *The composite operator acting on trajectories is*

$$G = U \circ F \circ K \circ C,$$

where C performs localized orthogonal compression, K applies curvature intensification/clipping, F induces loss of injectivity (folding), and U realises stabilisation (unfolding). All four

maps preserve the Riemannian structure and satisfy the contraction properties proved in Zenodo 10.5281/zenodo.19210137.

3 Fruit-Fly Connectome Toy Model

We consider a minimal toy model inspired by the FlyWire connectome. Let each neural cluster $i = 1, \dots, N$ be represented by a 2-D state vector (x_i, y_i) . The collective state evolves under the coupled map

$$\begin{pmatrix} x_i^{t+1} \\ y_i^{t+1} \end{pmatrix} = g_{b,\alpha} \left(G(x_i^t, y_i^t) \right) + \text{weak synaptic noise}.$$

After exactly six iterations of G the numerical trajectories converge to a low-dimensional attractor whose topology is consistent with observed connectome modules.

```

1  # Minimal Fruit-Fly Toy Model (Python pseudocode)
2  def G(state, params):
3      # C: localized orthogonal compression (5x5 patch projection)
4      # K: curvature clipping with bias b
5      # F: folding (local averaging)
6      # U: unfolding (stabilisation step)
7      return unfold(fold(konst(comp(state))))
8
9  for t in range(6):
10     swarm = [G(agent, params) for agent in swarm]
```

4 Saturated Pitchfork Bifurcation in Biological Context

The 2-D coupled map $g_{b,\alpha}$ undergoes a saturated pitchfork bifurcation at $|\alpha| = 0.5$. In the fruit-fly toy model this corresponds to the transition between a single stable rest state and a pair of bounded oscillatory states (HPA-axis analogue). All statements follow directly from the theorems proved in Zenodo 10.5281/zenodo.19210137.

5 AXLE Formalization Module (Lean 4)

The following Lean 4 structures and theorems formalise the biological multi-agent realisation. The complete module lives in the public AXLE repository:

<https://github.com/TOTOGT/AXLE>

```

1  structure BioAgent where
2    trajectory : List Real
3    coupling   : Real
4
5  def BioSwarm := List BioAgent
6
7  theorem swarm_contraction (swarm : BioSwarm) (alpha : Real)
8    (h : |alpha| < 0.5) :
9    LipschitzWith (0.5 + |alpha|) (fun s => List.map applyG s) := sorry
10
11 theorem collective_fixed_point (swarm : BioSwarm) :
12   Exists (fun eq => List.map (fun a => g_b a.coupling a.trajectory.head
13     !) swarm = eq) := sorry
14
15 theorem bio_pitchfork (alpha : Real) (h : |alpha| = 0.5) :
16   HasSaturatedPitchfork (fun s => g_{b,alpha} s) := sorry
```

6 Discussion

The Fruit-Fly Connectome Toy Model demonstrates that the operator pipeline $G = U \circ F \circ K \circ C$ and its associated fixed-point and bifurcation theorems provide a coherent mathematical language for biological transitions. No new invariants are claimed; the dm^3 triple is used only for normalisation. All code and formal proofs are maintained in the AXLE repository at <https://github.com/TOTOGT/AXLE>.